The Information Coefficient (IC)

Measuring the Predictive Power of a Signal

IEOR E4576 — Data-Driven Models in Finance — Fall 2026 — Week 4

1 Definition

The **information coefficient** is the correlation between a signal f_t and the subsequent realised return $r_{t,\tau}$ over forward horizon τ:

$$\mathrm{IC}(\tau) = \mathrm{Corr}\big(f_t, r_{t,\tau}\big)$$

Corr denotes Pearson (linear) or Spearman (rank) correlation; the Spearman variant is standard in high-frequency work (Section 4).

2 Return Measurement

For daily close-to-close data, minute bars, and other bar-based data, the price is the **last trade** in each interval (the close), and returns are measured close to close:

$$r_{t,\tau} = \frac{P_{t+\tau} - P_t}{P_t},$$

where P_t is the last trade price in interval t. At these horizons the typical return is large relative to the bid–ask spread, so the noise from trading at the bid or the ask is small.

At very high frequency the spread is comparable to the price moves themselves. There, returns are measured **mid-to-mid**, using $m_t = \frac{1}{2}(\mathrm{bid}_t + \mathrm{ask}_t)$ in place of P_t: last-trade prices bounce between bid and ask, which adds negative autocorrelation that can make a null signal appear predictive, or mask a genuine one.

3 Interpretation and Practical Thresholds

IC = 0 means no predictive power; |IC| = 1 means perfect prediction.

|IC|	Interpretation
< 0.02	Noise; no evidence of predictive power
0.02–0.05	Weak; possible edge, requires a large sample to confirm
> 0.05	Meaningful at high frequency; worth further development
> 0.10	Strong; rare at high frequency

4 Pearson vs. Spearman IC

- **Pearson IC** operates on raw values and measures linear association.
- **Spearman IC** replaces each value with its rank, then applies the Pearson formula, capturing any monotonic relationship.

Spearman is preferred at high frequency because order-flow signals are heavy-tailed — a single outlier burst can dominate a Pearson estimate — while ranking is robust to extremes.

5 Statistical Testing

A single IC estimate is noisy: its standard error over N observations is $\approx 1/\sqrt{N}$, giving $t \approx \mathrm{IC}\sqrt{N}$. The **ICIR** — the mean IC divided by its time-series standard deviation — measures consistency and is a more reliable guide than any single estimate.